

ECONOMICS

Correcting for spatial and temporal dependence in estimating the economic effect of extreme heat

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Callahan and Mankin argue that extreme heat, measured by maximum 5-day average temperature, has a precise and economically meaningful effect on subnational gross domestic product per-capita growth. We reexamine this claim and find that the reported precision does not survive methods that respect the data's spatial and temporal dependence. Country-preserving permutations and postselection correction substantially widen marginal-effect intervals. The estimates are also highly sensitive to influential countries and years, with exclusions sometimes collapsing the effect or reversing its sign. A hierarchical Bayesian model with region-within-country structure and AR(1) year effects yields much larger uncertainty, shrinks the estimated effect by 67 to 90%, and produces credible intervals spanning zero. Last, rolling out-of-sample tests show that adding extreme-climate variables does not improve prediction over models without them. We conclude that the claimed precision is not sustained once dependence, model selection, and influential observations are handled appropriately.

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INTRODUCTION

Accurately quantifying the economic costs of global warming is fundamental and a prerequisite for any effective climate policy. Callahan and Mankin (1) (henceforth C&M) contribute to this urgent task by statistically modeling the relationship between extreme heat and economic output, reporting statistically significant damage estimates that diverge sharply across regions. These estimates are incorporated into a larger causal chain in Callahan and Mankin (2) as shown in Fig. 1.

Because this claim sits at the intersection of climate science, econometrics, and policy, its credibility is central to the broader project of linking climate variability to economic outcomes. If valid, their framework would substantially strengthen the evidentiary basis for estimating the economic damages of climate change and could inform both macroeconomic projections and emerging legal and policy debates over climate liability.

Using C&M's own data from Callahan and Mankin (1, 2) and their replication materials, we reexamine the regression link of this causal chain—the third arrow in Fig. 1, which maps temperature and extreme-heat variation to changes in economic growth—and find that the precision attached to its reported coefficients does not survive standard diagnostics for data with strong spatial and temporal dependence. Our analysis bears on this regression link only, not on the climate-model attribution step that precedes it in Callahan and Mankin (2). C&M's analysis understates statistical uncertainty by ignoring the strong spatial and temporal correlation of regional growth, paralleling the core error identified by Schötz (3) in his critique of Kotz *et al.* (4). Once the dependence structure is taken seriously, the apparent precision of C&M's reported SEs cannot be sustained, and the data are consistent with extreme-heat effects ranging from zero to moderate.

At the same time, economic theory helps explain why such uncertainty is not merely a statistical artifact. The effect of extreme heat on measured gross domestic product (GDP) growth is likely to operate through multiple channels, including labor productivity, agricultural output, energy demand, health, infrastructure disruption, firm

adaptation, migration, and public or private investment responses (5). These channels may differ sharply across sectors, regions, income levels, institutions, and time horizons. Some mechanisms imply immediate output losses, while others imply substitution, adaptation, or delayed effects. As a result, the aggregate relationship between a short-run extreme-heat measure and subnational GDP per-capita (GDPpc) growth need not be stable, homogeneous, or easily summarized by a single precisely estimated marginal effect. In economic terms, the data reflect many potentially offsetting mechanisms, differing margins of adaptation, and heterogeneous institutional settings, all embedded in strong regional dependencies. This theoretical heterogeneity reinforces our empirical findings: Pronounced spatial and temporal dependence means that the nominally large panel contains substantially less independent information than its raw number of observations suggests.

The core statistical difficulty is to capture the effect of the dependence structure on the SEs. Subnational economic growth exhibits pronounced dependence: Regions within a country comove within a year, growth is serially correlated across years, and climate variables themselves are strongly spatially autocorrelated. Under such conditions, standard regression assumptions are violated: Regression residuals are neither independent nor identically distributed. C&M attempt to address this by reporting Huber-White SEs clustered by region, but clustering only by region cannot account for contemporaneous country-year shocks or for shared year effects. As we demonstrate below, this mis-specification materially understates uncertainty, undermining the robustness of their headline results and illustrating why conventional and “robust” SEs are insufficient in this setting. The concern is recognized in this literature: Hsiang (5) writes that “uncertainty estimates for regressions must account for the strong spatial and temporal autocorrelation in climatic exposure, regardless of the research design employed.” Our analysis operationalizes the kind of correction Hsiang has in mind in the C&M setting.

Before presenting our reanalysis, we briefly summarize C&M's original model and data. The authors regress regional GDPpc growth on mean temperature, $Tx5d$, their interaction, temperature variability and its interaction with the annual temperature cycle, and precipitation, including region and year fixed effects. Their marginal effect gives the change in growth associated with a 1°C increase in $Tx5d$.

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Fig. 1. Causal chain from greenhouse gas increases to economic damages.

Their standardized marginal effect scales this quantity by the average within-region SD of $Tx5d$ (approximately 1.3°C). After excluding numerous observations with missing growth or missing climate variables, the training dataset contains 27,283 region-year observations from 76 countries and 1537 regions. Because missingness is disproportionately concentrated in poorer and hotter parts of the world, sample selection introduces an additional concern. Using the released replication files, we were able to reproduce their main reported coefficients and figures.

Our reanalysis proceeds in four parts. First, we evaluate the sensitivity of C&M's estimates to alternative inference procedures that are designed for data with strong spatial and temporal dependence, particularly dependence-respecting permutation tests. Second, we examine the influence of potentially outlying countries and years. Third, we fit a hierarchical Bayesian model that incorporates spatial clustering and temporal autocorrelation, aligning the model structure with the data-generating process. Fourth, we evaluate out-of-sample predictive performance using blocked cross-validation and rolling one-step-ahead forecasts. Across these exercises, we find that the analytic margins of error C&M attached to the headline marginal effect substantially widen once the dependence structure is properly accounted for. The estimates are also highly sensitive to a small number of influential countries and years, and including climate variables produces essentially no out-of-sample gain over baselines that use none.

Together, these results suggest that the apparent precision of C&M's estimates is overstated by an inference procedure that does not adequately account for the spatial and temporal dependence in the panel. The case illustrates the importance of addressing such dependence when estimating macroeconomic climate damages and calls for caution in treating these regression estimates as a precise foundation for downstream attribution or policy work.

Our analysis proceeds in four parts, each showing that the reported precision of the $Tx5d$ marginal effect does not survive when the analysis properly accounts for dependence, influential observations, and predictive performance. Together, these results show that C&M's reported SEs substantially understate the true uncertainty about the effect of extreme heat on regional growth, while leaving open a wide range of plausible effect sizes from zero to moderate.

1) A dependence-respecting permutation test reveals that the analytic intervals are too narrow. C&M's placebo tests randomize $Tx5d$ across the full sample, within years, or within regions, but these permutations destroy the very spatiotemporal dependencies that generate spurious significance, so they are too weak to diagnose it. We instead implement a country-year-preserving permutation: for each country, we randomly permute the years and apply that same permutation to all regions in the country, thereby eliminating any true $Tx5d$ -growth relationship while preserving the dependence structure capable of producing false positives. Because $Tx5d$ and mean temperature T are correlated, we jointly permute them. Under this permutation, the null distributions are far wider than the analytic intervals C&M report and routinely generate large positive and negative marginal effects even though the true effect is zero by

construction. A postselection permutation—mirroring C&M's choice of $Tx5d$ from six competing heat metrics—raises the false-discovery probability by an additional 5 to 10 percentage points (p.p.). Once the dependence structure and metric selection are respected, the analytic confidence intervals reported in C&M understate the true range of marginal effects compatible with the data.

2) The reported effects hinge on a few highly influential outliers. Large, spatially correlated panels are acutely vulnerable to influential years and countries. When we assess influence at the year and country level (rather than merely at the region-year level), we find that removing 1998, 2009, Mongolia, and Kenya reduces the estimated $Tx5d$ effect at $T = 5^{\circ}\text{C}$ by 79% ($0.764 \rightarrow 0.160$ p.p. per SD) and reverses the sign at $T = 25^{\circ}\text{C}$ ($-0.892 \rightarrow +0.124$). Such marked collapses from modest deletions show that the original estimates are not robust but precariously balanced on a handful of outlying observations. The reported uncertainty is far too small to be credible.

3) A hierarchical Bayesian model that respects spatial and temporal dependence finds effects close to zero with much wider uncertainty. To bring the model into alignment with the data-generating structure, we fit a Bayesian model with region effects nested within countries and AR(1) year effects, using weakly informative priors on climate coefficients. The resulting posterior intervals are wide and centered near zero, with point estimates shrinking by roughly 67% at $T = 5^{\circ}\text{C}$ and nearly 90% at $T = 25^{\circ}\text{C}$ relative to C&M's values. These results make clear that the narrow margins of error C&M attach to their headline estimates cannot be sustained once the model acknowledges spatial clustering and temporal autocorrelation.

4) The fitted regression has limited out-of-sample explanatory value for region-year growth. A regression used as the input to a damage chain should explain held-out data better than a baseline that uses no climate signal; the C&M specification does not. First, the in-sample R^2 of C&M's model is 0.0946, only marginally higher than the same model without climate variables ($R^2 = 0.0883$). Second, both i.i.d. 10-fold cross-validation and blocked (contiguous-year) cross-validation show that including climate variables does not improve predictive accuracy. Third, even a rolling one-step-ahead forecast using our Bayesian model, which benefits from its AR(1) year structure, shows essentially zero median reduction in forecast error when climate covariates are included. This does not, by itself, rule out a small but real causal effect; it does indicate that the fitted regression contains limited stable signal beyond what region and year fixed effects already capture.

Together, these four lines of evidence show that the precision claimed in C&M's reported SEs is not robust: Once spatial and temporal dependence are properly accounted for, the data are consistent with extreme-heat effects ranging from approximately zero to moderate, rather than the precisely estimated effect reported in C&M.

Implications and contribution

Our analysis makes two broad points. Substantively, using C&M's own data, the precision attached to the C&M estimates cannot be sustained once spatial and temporal correlation, metric selection, and outliers are properly addressed; the data are consistent with

effects ranging from approximately zero to moderate, but they do not support the narrow margins of error C&M report. Methodologically, the case illustrates why climate-economy panel regressions with a strong dependence structure require designs that go beyond region-clustered SEs: Valid uncertainty quantification requires dependence-respecting resampling or a fully specified hierarchical time-series structure. Together, these analyses summarize the central message: What looks like a sharp, precisely estimated effect under the original specification becomes small and uncertain under methods that respect the data's dependence structure. Nonetheless, policy-makers need not wait for a single precise global damage estimate to act. Instead, policy analyses should propagate the wider uncertainty and test whether decisions remain robust across plausible heat-growth effects that vary by place, sector, and adaptation capacity.

Roadmap

The remainder of the paper is organized as follows. The “Background and original model” section reviews and replicates C&M's original model and dataset. The “Permutation tests” section presents our dependence-respecting permutation tests, including the postselection variant. The “Outliers” section examines sensitivity to influential countries and years. The “Bayesian model” section develops the hierarchical Bayesian model and reports posterior estimates. The “Validating the model: out-of-sample prediction tests” section evaluates predictive performance using rolling forecasts and cross-validation.

RESULTS AND DISCUSSION

Background and original model

Callahan and Mankin (1, 2) study the effect of extreme heat on subnational economic growth using an econometric “damage-function” framework. They combine climate data with regional GDPpc time series to estimate how growth responds to annual mean temperature, the intensity of extreme heat events, and other climate variables. Their framework lies in the panel fixed-effects tradition of Dell *et al.* (6) and Burke *et al.* (7). Their primary measure of extreme heat is $Tx5d$, the maximum 5-day average temperature in a region-year. The central objective is to estimate the effect of $Tx5d$ on economic growth.

Model

Their primary regression model relates annual GDPpc growth g_{it} in subnational region i and year t to region fixed effects $\alpha_i^{(\text{region})}$, year fixed effects $\alpha_t^{(\text{year})}$, and several climate covariates: annual mean temperature T_{it} , extreme heat $Tx5d_{it}$, temperature variability V_{it} (computed as the average, across months in year t , of the within-month SD of daily mean temperatures for region i), the region-specific annual temperature cycle A_i (computed for each region i as the average, across all years, of the within-year difference between the warmest and coldest monthly mean temperatures), and annual accumulated precipitation P_{it} . Formally

$$g_{it} = \beta_1 T_{it} + \beta_2 T_{it}^2 + \beta_3 Tx5d_{it} + \beta_4 (Tx5d_{it} \times T_{it}) + \beta_5 V_{it} + \beta_6 (V_{it} \times A_i) + \beta_7 P_{it} + \alpha_i^{(\text{region})} + \alpha_t^{(\text{year})} + \varepsilon_{it} \quad (1)$$

To address heteroskedasticity and within-region serial dependence, they report Huber-White SEs clustered by region; this allows arbitrary correlation within a region across time but assumes independence across regions.

Callahan and Mankin (1, 2) focus on the marginal effect of extreme heat, $\partial g / \partial Tx5d = \beta_3 + \beta_4 T$, which gives the change in growth (p.p.) associated with a 1°C increase in $Tx5d$. Because within-region variability in $Tx5d$ differs across region, they scale this quantity to obtain a standardized marginal effect

$$ME(T) := (\beta_3 + \beta_4 T) \sigma_{Tx5d} \quad (2)$$

where $\sigma_{Tx5d} \approx 1.3^\circ\text{C}$ is the average within-region SD of $Tx5d$ (computed as the average across regions of the within-region SD of $Tx5d$). This quantity is reported in p.p. of GDPpc growth for a 1 SD increase in extreme heat intensity.

Data

Callahan and Mankin's dataset spans 38 years (1979 to 2016) and initially contains 137,180 region-year observations from 228 countries and 3610 regions. Of these, 107,189 observations lack GDPpc-growth data, reflecting limited availability of subnational economic statistics in much of the world. Dropping these yields 29,991 observations from 76 countries and 1537 regions. The excluded regions are disproportionately located in low-income tropical countries—47 in Sub-Saharan Africa and 30 in Latin America and the Caribbean—raising concerns about sample representativeness: the fitted model is estimated primarily on relatively cooler and wealthier regions while omitting many of the hottest and poorest, which are most vulnerable to extreme heat. After additionally removing rows with missing climate covariates, the final training sample for model (1) contains 27,283 observations.

Replication

We successfully replicated their primary regression results. In particular, we exactly reproduced table S1 of Callahan and Mankin (1), which reports the regression coefficients and clustered SEs for model (1). We also closely reproduced their figure S2 (discussed in the “Permutation tests” section), which displays the marginal effects of extreme heat alongside their placebo/permutation test results.

Permutation tests

Callahan and Mankin (1) implement a set of permutation, or “placebo,” tests to assess whether the apparent significance of extreme-heat effects reflects a genuine relationship, given that strong autocorrelation in both climate and economic series can produce spurious significance. They repeatedly reestimate model (1) after permuting $Tx5d$ under three schemes: (i) across the full sample, (ii) within years, and (iii) within regions. In each case, the estimated effects vanish under randomization (see their figure S2), which is a necessary, though not sufficient, condition for interpreting the $Tx5d$ effect as reflecting a real underlying relationship.

We reproduce their figure S2 in the first four paired intervals of Fig. 2. The first two pairs (labeled “Original model”) show the mean and 95% confidence intervals of the standardized marginal effect (2) from the original model (1), and these closely match the corresponding results in Callahan and Mankin (1). Our intervals are not identical to theirs; section S1 details our computation of SEs for the marginal effects. The other three pairs of intervals correspond to their three permutation tests. As in the original study, the estimated effects vanish under randomization, which, to Callahan and Mankin, indicates that the signal they detect in the unpermuted data is not an artifact of chance alignment in $Tx5d$ but instead reflects a meaningful relationship between extreme heat and growth.

However, the permutation schemes in Callahan and Mankin (1) disrupt the full spatiotemporal correlation structure of the data

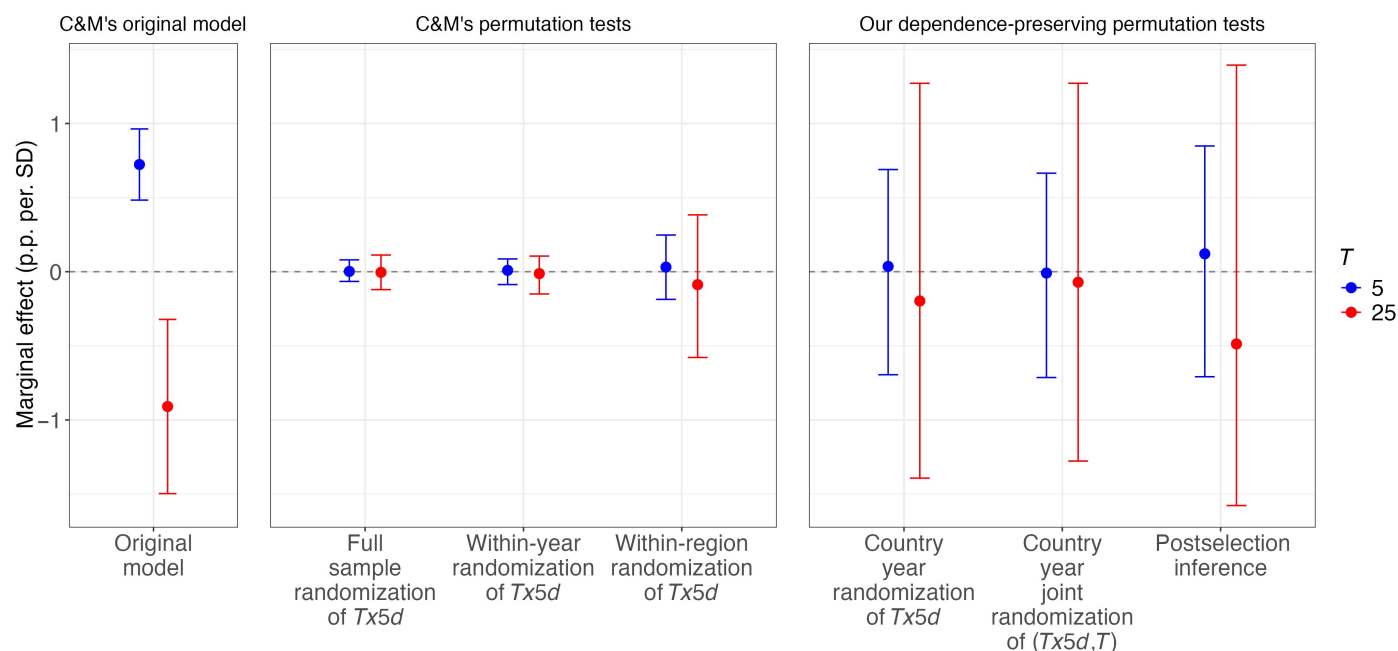


Fig. 2. Marginal effects of extreme temperature $Tx5d$ on economic growth g when the annual average temperature is $T = 5^\circ\text{C}$ (blue) and $T = 25^\circ\text{C}$ (red). The first pair of intervals show the mean and 95% confidence intervals of the standardized marginal effect (2) from the original model (1). The remaining pairs of intervals show the median and 95% intervals for the various permutation tests.

(illustrated in Fig. 3). Preserving that structure is essential for diagnosing spurious significance. Within a country-year, regional growth outcomes are tightly clustered, and growth dynamics differ markedly across countries.

Because the original randomization schemes disrupt key elements of the spatiotemporal dependence structure—treating all observations as exchangeable in the full-sample randomization and ignoring country-level clustering in the within-year and within-region randomizations—they are not strong enough to probe the primary source of spurious correlation. To address this, we construct a permutation procedure that preserves the joint country-year structure of the panel. For each country, we draw a uniformly random permutation of the years and apply the same reordering of $Tx5d$ to all regions within that country. Intuitively, this scrambling does two things at once. It destroys the true climate signal, because year t growth is now paired with a $Tx5d$ value drawn from a different year, so any real climate-growth link is broken by construction; and it leaves the panel's dependence structure intact: Growth itself is untouched, so its within-year clustering across regions and its year-to-year autocorrelation are unchanged; and because the same year permutation is applied to every region within a country, regions within a country still move together in $Tx5d$, preserving the within-country spatial comovement of the climate variable as well. A real climate signal should therefore be erased by this scrambling, whereas a dependence-induced spurious effect should not be. We also implement a second variant that jointly permutes $Tx5d$ and T , because these two variables are themselves correlated.

The results of these two dependence-preserving permutation tests—the country-year permutation of $Tx5d$ and of $(Tx5d, T)$ —appear in the fifth and sixth slots of Fig. 2. These schemes produce very wide intervals for the marginal effects. Even though growth and

extreme temperature have no relationship by construction, the model frequently produces large positive or negative marginal effects depending on the permutation. The corresponding extreme-temperature coefficients are statistically significant in roughly half of the permutations (see fig. S2). These results demonstrate that a panel with strong spatial correlation and temporal autocorrelation can readily produce spurious marginal effects. They further show that, by not fully accounting for this spatiotemporal dependence, model (1) substantially understates the true uncertainty in both its parameters and in the estimated effect of extreme heat on economic growth.

It may be tempting to say the country-year permutation simply has less power than the region-year scheme, but this lower power is a consequence of retaining the spatial dependence among regions within a country and the serial dependence in regional outcomes more faithfully—a fidelity that necessarily removes much of the within-country temporal variation that the region-year test treats as independently informative. This loss of power is not an incidental weakness of the test; it is the inferential cost of no longer treating dependent observations as independent. The model-based margins of error from the dependence-adjusted specifications in the “Bayesian model” section are roughly twice as large as those from C&M's original specification, which corresponds to an effective sample size about four times smaller than C&M's analytic inference assumes (because SEs scale as the inverse square root of the effective sample size). Nonrejection under our permutation should therefore be read as further evidence that C&M's analytic inference rests on an effective sample size that is too large once spatial and temporal dependence are taken seriously.

Postselection inference

Callahan and Mankin (1) write that they use $Tx5d$ as their measure of extreme heat because “the peak 5-day period yields the strongest

Growth over time by region within each country

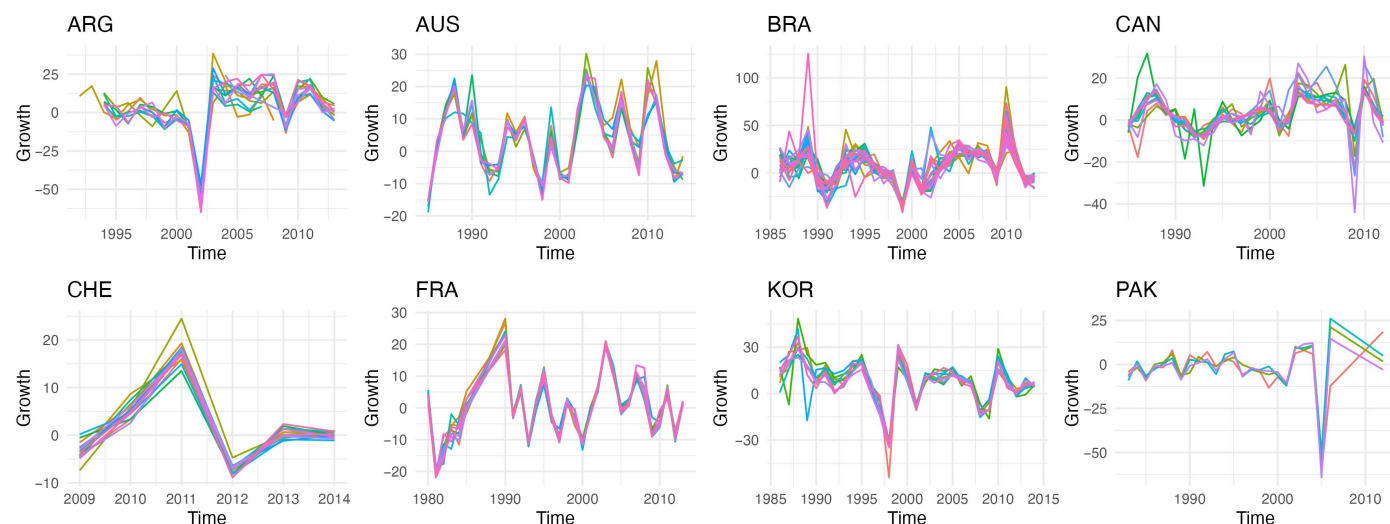


Fig. 3. Economic growth (y axis) versus time (x axis) for various countries (subplot) and each region (color) in those countries.

response,” whereas other measures yield “weaker and more uncertain responses” (see their figure S3). They chose *Tx5d* (the maximum 5-day average temperature in each region and year) over five other measures of extreme heat—*Txx* (hottest day), *Tx3d* (3-day), *Tx7d* (7-day), *Tx15d* (15-day), and *Tmonx* (hottest month)—because the magnitude of its effect size was the largest. It is important to account for this selection step because choosing among multiple extreme-temperature measures and reporting the one with the most favorable effect risks overstating evidence. By chance alone, some metrics will yield stronger or more precise results even in the absence of a true relationship. Permutation tests that ignore this model selection therefore understate the probability of obtaining spurious marginal effects.

Hence, we implement an additional “postselection inference” permutation test. As before, for each country, we draw a uniformly random permutation of the years and jointly permute all six extreme-temperature measures (*Txx*, *Tx3d*, *Tx5d*, *Tx7d*, *Tx15d*, and *Tmonx*) for every region. For each permutation, we fit one model per measure and compute six marginal effects. We then select the measure producing the most extreme effects, specifically maximizing $\max\{ME(T=5), 0\} - \min\{ME(T=25), 0\}$ so that the marginal effect at $T=5$ is as positive as possible while that at $T=25$ is as negative as possible—the natural target because C&M present their headline result as $ME(T)$ at these two endpoints.

The results of this postselection test appear in the rightmost slot of Fig. 2. The selected marginal effects are, on average, more extreme than in the baseline dependence-preserving tests, though the permutation intervals remain wide. Also, the resulting permutation intervals fully eclipse the original-model intervals, implying that the range of effects produced under random permutations is substantially wider than the range implied by model (1). Further, the share of permutations in which the extreme-temperature coefficients are statistically significant increases by roughly 5 to 10 p.p. (see fig. S2). As before, these findings show that spurious marginal effects arise easily under the strong spatiotemporal correlation structure of the

data and that model (1) materially understates the true uncertainty in its parameters.

Outliers

In the “Permutation tests” section, we showed that permutation tests that better preserve the spatiotemporal correlation structure frequently produce spurious strong marginal effects, suggesting that the analytic uncertainty in model (1) substantially underestimates uncertainty surrounding the effect of extreme heat on economic growth. In this section, we present a second line of evidence that the original analysis understates uncertainty: The estimated marginal effect of *Tx5d* is highly sensitive to the inclusion or exclusion of a small number of individual countries and years. Neither Callahan and Mankin (1) nor Callahan and Mankin (2) includes such an analysis, even though the strong spatiotemporal correlation structure of the panel (recall Fig. 3) makes country- and year-level influence an important unit of analysis beyond just the individual region-year observation.

A recent *Washington Post* article (8) underscores this concern in our same context of estimating climate damages. It reports on Kotz *et al.* (4), a study similar in spirit to Callahan and Mankin’s, which projected global GDP losses of 19% by 2050 and 62% by 2100 under a high-emissions scenario. The study attracted wide attention, being the second-most cited climate paper in the media in 2024 (9) and informing analyses by the US government, the World Bank, and other institutions. Yet, Bearpark *et al.* (10) showed that removing a single country—Uzbekistan—markedly altered the results, reducing the projected losses from 19 to 6% by 2050, and from 62 to 23% by 2100. It should be noted, however, that this was due to leverage arising from a documented data error in the Uzbekistan series—unlike the leverage in the C&M panel documented below, which is not attributable to a data defect. Nonetheless, the similarity remains: A global panel coefficient reported with narrow CIs depends critically on just a few units, signaling substantially understated uncertainty.

This example illustrates how sensitive large-scale climate-economy estimates can be to outliers, and highlights the importance of assessing the influence of entire countries or years. Our own analysis reveals similar sensitivity: A small number of countries and years exert disproportionate influence on the estimated marginal effects of $T \times 5d$. Removing just Mongolia and Kenya reduces the marginal effect at $T = 5$ from 0.764 to 0.500 (a 35% decline) and moves the effect at $T = 25$ from -0.892 to -0.287 (a 68% shift toward zero). Removing just 1998 and 2009 yields even larger changes: The marginal effect at $T = 5$ falls from 0.764 to 0.361 (a 53% decline), and the effect at $T = 25$ shifts from -0.892 to -0.171 (an 81% shift toward zero). Removing these four outliers together—Mongolia, Kenya, 1998, and 2009—drops the $T = 5$ marginal effect from 0.764 to 0.160 (a 79% decline), and the $T = 25$ marginal effect reverses sign entirely, from -0.892 to 0.124 (a 114% increase). That the estimated effects change so markedly when only a few years or countries are removed indicates that the original estimates are highly sensitive to outliers and that the reported uncertainty is likely substantially underestimated.

To assess this sensitivity systematically rather than at hand-selected units, we supplement the deletions above with a full leave-one-out jackknife of model (1) at the country and year levels (Figs. 4 and 5). For each country (year), we drop that country (year) from the panel, refit the model, and record the change in the global standardized marginal effect $ME(T)$, expressed in z -score units of C&M's analytic SE for that marginal effect (so a shift of magnitude 1 means a 1 SE move). If those SEs were a faithful description of sampling uncertainty for the global marginal effect, no single country or year should be able to move that effect by more than a small fraction of one such SE.

Many countries and years move it substantially: Removing Mongolia shifts $ME(T=5)$ by roughly -1.8 analytic SEs, removing Turkey shifts it by $+1.1$, and removing Kenya shifts $ME(T=25)$ by $+1.1$;

Brazil, Mexico, Honduras, Indonesia, Ireland, Kazakhstan, Russia, Sweden, France, Ukraine, Romania, and Canada each move at least one of the two marginal effects by 0.4 to 0.7 SE; at the year level, 2003 and 2006 each move one of the marginal effects by more than a full analytic SE, in addition to 1998 and 2009. The C&M global marginal effect is thus quite sensitive to a small set of observations, providing further evidence that C&M understated uncertainty. A similar conclusion has now come from the authors of Kotz *et al.* (4) themselves, in the wake of Bearpark *et al.* (10) and Schötz (3): “the original assessments of uncertainty were too small.”

Bayesian model

The permutation tests of the “Permutation tests” section and the outlier analysis of the “Outliers” section indicate that the original analysis in the “Background and original model” section understates uncertainty. In particular, the original specification does not adequately capture the spatiotemporal correlation structure in the data (Fig. 3). In this section, we develop a Bayesian version of model (Eq. 1) that more appropriately reflects this structure: In effect, we build the dependence visible in the data—regions clustered within countries, growth correlated across adjacent years—into the model itself, rather than assuming it away.

Ideally, one would allow for full space-time interactions (e.g., country $\times$ year coefficients), because Fig. 3 shows that growth dynamics vary across countries and years. However, such a specification would inflate uncertainty to a prohibitive level and risk obscuring the more fundamental issues in Callahan and Mankin's approach. We therefore adopt a model that mirrors the original specification while incorporating hierarchical spatial structure and temporal autocorrelation.

The likelihood is

$$g_{it} = \beta_0 + \eta_{it} + \alpha_i^{(\text{region})} + \alpha_t^{(\text{year})} + \varepsilon_{it} \quad (3)$$

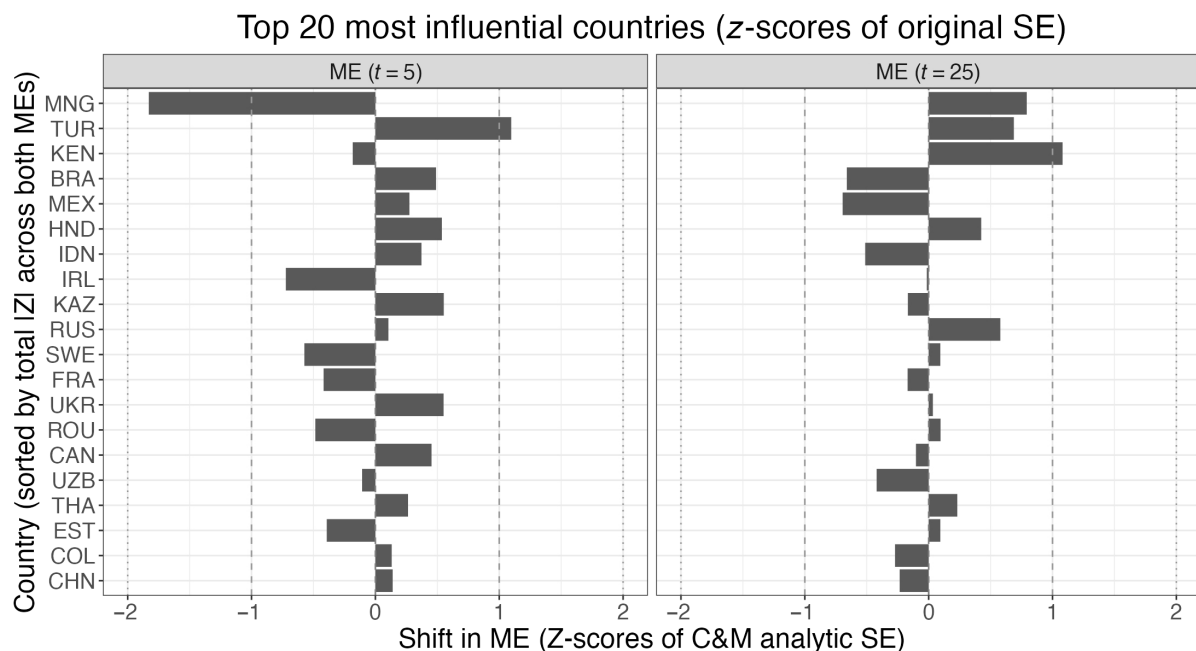


Fig. 4. Top 20 most influential countries in the leave-one-country-out jackknife of the primary Bayesian model. Each bar gives the shift in the global standardized marginal effect $ME(T)$ when the corresponding country is dropped, in units of C&M's analytic SE for that marginal effect. Dashed lines: ± 1 SE; dotted lines: ± 2 SE.

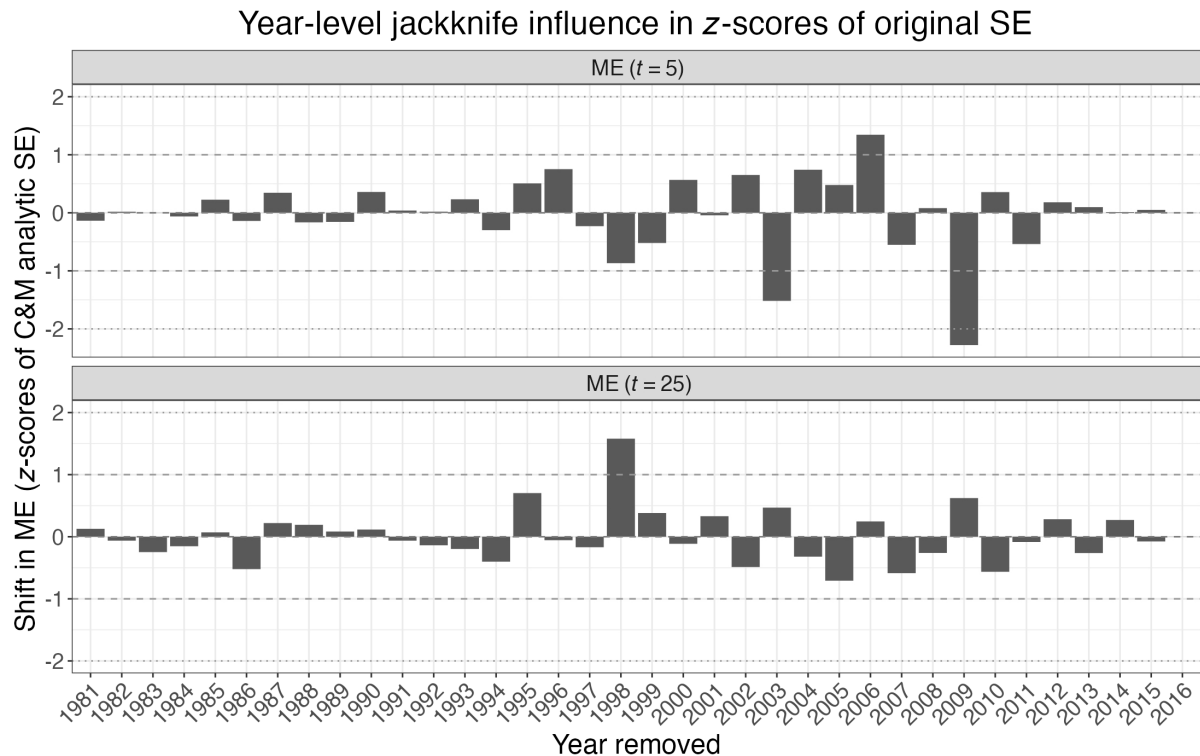


Fig. 5. Year-level jackknife influence on the global standardized marginal effects of C&M's original model. Dashed lines: ± 1 SE; dotted lines: ± 2 SE.

where η_{it} is a linear function of the climate covariates

$$\eta_{it} = \beta_1 T_{it} + \beta_2 T_{it}^2 + \beta_3 Tx5d_{it} + \beta_4 (Tx5d_{it} \times T_{it}) + \beta_5 V_{it} + \beta_6 (V_{it} \times A_i) + \beta_7 P_{it} \quad (4)$$

We assign weakly informative priors for the intercept and climate coefficients, $\beta_* \sim \mathcal{N}(0, 10^2)$.

To capture spatial clustering, we model regional effects hierarchically

$$\alpha_i^{(\text{region})} \sim \mathcal{N}[\alpha_i^{(\text{country})}, \sigma^2(\text{region})] \quad (5)$$

$$\alpha_i^{(\text{country})} \sim \mathcal{N}[0, \sigma^2(\text{country})] \quad (6)$$

To capture temporal autocorrelation, year effects follow an AR(1) process (growth is correlated from year to year, particularly at lag-1; see fig. S3)

$$\alpha_t^{(\text{year})} \sim \mathcal{N}[\phi \alpha_{t-1}^{(\text{year})}, \sigma^2(\text{year})] \quad (7)$$

Observation errors are i.i.d. $\varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$. Variance parameters σ^{2*} receive half-normal priors $\mathcal{N}_+(0, 2^2)$, and the autoregressive coefficient has prior $\phi \sim \mathcal{N}(0, 0.5^2)$ truncated to $[-1, 1]$.

Because the posterior distribution is not analytically tractable, we sample from it using MCMC implemented in Stan (11) via the rstan package (12). We run four chains for 20,000 iterations each, discarding the first 10,000 as warm-up and retaining every 10th post-warm-up

draw, yielding 4000 posterior draws. Additional details on the fitted model appear in section S3.

Figure 6 shows the posterior mean and 95% credible intervals of the marginal effects (Eq. 2) from this Bayesian model. Uncertainty is much larger than in the original model (Fig. 2), and the estimated effects shrink substantially toward zero: at $T = 25^\circ\text{C}$, by about 90%, and at $T = 5^\circ\text{C}$, by about 67%. The posterior is consistent with either a null or modest effect, underscoring the high degree of uncertainty in this setting.

We note that the credible interval spanning zero is a posterior summary conditional on the model structure, not a model-free test. Sensitivity checks on the year-effect specification and the prior scale (fig. S5) show that this conclusion is stable across plausible alternatives—Neither the AR(1) structure nor the choice of prior scale is what drives the zero-spanning interval; the data do.

Lag growth model

As an alternative specification for the temporal nuisance structure, we also fit a Bayesian model identical to the specification above except that the AR(1) year effect is replaced by lag growth $\beta \cdot g_{i,t-1}$, where $g_{i,t-1}$ is regional growth in the previous year. The motivation is direct: Growth itself is strongly autocorrelated, and a lag of the dependent variable absorbs that autocorrelation more sharply than a smooth year-level AR(1) does. In-sample, replacing the year effect with lag growth raises R^2 by 15%; with the outliers Mongolia, Kenya, 1998, and 2009 removed, the improvement is 19%.

The lag-growth specification yields qualitatively different conclusions from the AR(1) year-effect specification (see Fig. 7). The marginal-effect posteriors shift upward, and, notably, the marginal effect of extreme heat at $T = 25^\circ\text{C}$ flips sign: Under C&M's original specification

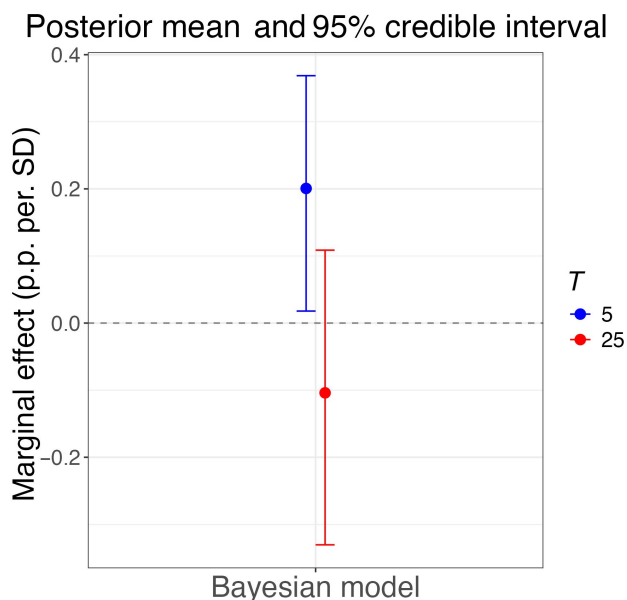


Fig. 6. Posterior mean and 95% credible intervals from the primary Bayesian model for the marginal effects of extreme temperature $Tx5d$ on economic growth g at annual average temperatures of $T = 5^\circ\text{C}$ (blue) and $T = 25^\circ\text{C}$ (red).

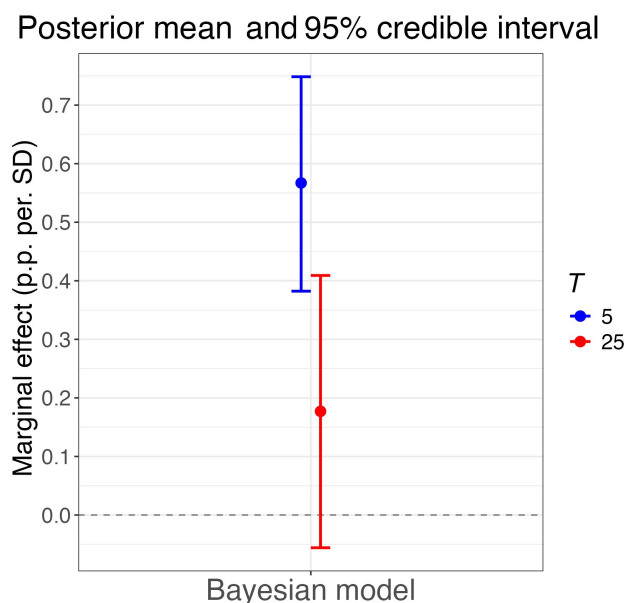


Fig. 7. Posterior mean and 95% credible intervals from the Bayesian model with lag growth $\beta \cdot g_{i,t-1}$ replacing the AR(1) year effect, for the marginal effects of $Tx5d$ on growth at $T = 5^\circ\text{C}$ (blue) and $T = 25^\circ\text{C}$ (red).

[and under our AR(1) year-effect Bayesian model], the marginal effect at $T = 25$ is negative; under the lag-growth Bayesian model, it is positive. In other words, the lag-growth specification finds not estimated damages from extreme heat but estimated growth gains. The same pattern appears in the frequentist version: Replacing the year fixed effect in model (Eq. 1) with lag growth shifts the marginal effects upward and turns the extreme-heat marginal effect at $T = 25$ positive. Though combining a lagged dependent variable with unit

effects can produce finite-panel dynamic bias (13), such Nickell-type bias primarily concerns the unit effects and autoregressive parameter, both of which are nuisance parameters with respect to the climate effect of interest. Moreover, the bias is expected to be small with $T = 37$ postlag outcome years because it is of order $1/T$.

One could argue about which specification is “more correct.” We focus our main analysis on a model that retains a year effect because C&M’s specification uses a year effect, and our central goal is to ask whether their conclusions survive once spatial and temporal dependence are properly accounted for within their own modeling framework. However, the lag-growth specification is an equally plausible alternative, and it yields qualitatively different conclusions. That a small, defensible modification of C&M’s specification flips the sign of the headline marginal effect at the high-temperature endpoint is itself further evidence that the precision attached to C&M’s headline estimate cannot be sustained: The result is not robust to plausible variation in the dependence structure of the model.

Validating the model: Out-of-sample prediction tests

Our Bayesian model from the “Bayesian model” section better captures the spatiotemporal correlation structure and reflects much greater uncertainty in the marginal effects of extreme heat on economic growth. This substantial uncertainty, together with the considerable posterior mass on a null effect, raises a natural question: Should the climate variables be included in the model at all?

In their work, Callahan and Mankin treat model (Eq. 1) as causal. For instance, they assert that their model

“separates extreme heat from other factors that might affect economic growth, allowing [them] to isolate the causal effect of heat intensity on the hottest days of the year” (1).

This use of causal reflects a now-conventional usage in the panel fixed-effects econometric literature, but the standard statistical view holds that regression coefficients estimated from observational data are properly associational, and that the term causal is appropriate only for designs with explicit randomization, a credible source of exogenous variation, or a well-defended structural model (14). A two-way fixed-effects regression on a country-year panel of observational climate and growth data satisfies none of these; the fixed effects address specific forms of omitted-variable confounding but do not constitute identifying assumptions strong enough to license causal language without qualification. Which of these two views one adopts bears directly on how much weight to attach to the headline marginal effect. Beyond this disagreement, the statement above highlights a deeper issue: The model was not validated to determine whether it is useful—whether the climate variables should be included at all—let alone causal.

How should one validate that the climate variables contribute usefully to the fit? A standard diagnostic, following McShane and Wyner (15), is predictive performance: A model that cannot predict held-out data better than simple baselines provides little ground for treating its fitted coefficients as estimates of a stable underlying relationship. Our exercise is a direct extension to extreme heat of Newell *et al.* (16), who assessed the out-of-sample predictive accuracy of 800 GDP-temperature damage models and found no statistically significant temperature effects on GDP growth.

Weak out-of-sample performance does not, by itself, prove the absence of a real causal effect—A small but real effect may contribute little to region-year predictive accuracy when growth is dominated by nonclimate shocks. What it does indicate is that the fitted regression

contains limited stable signal beyond what region and year fixed effects already capture—further evidence, alongside the permutation tests of the “Permutation tests” section, the jackknife of the “Outliers” section, and the fitted Bayesian model of the “Bayesian model” section, that the precision C&M attach to the headline marginal effect is overstated.

As a starting point, we consider in-sample predictive performance. The R^2 of the original model (Eq. 1) with climate variables is 0.0946, only modestly higher (about 7%) than that of the same model without them ($R^2 = 0.0883$). These figures differ from the adjusted $R^2 = 0.186$ reported in figure S3 of Callahan and Mankin (1), but the two are not in conflict: Their 0.186 is the R^2 after the region and year fixed effects are partialled out, whereas the $R^2 = 0.0946$ we report measures the share of total variation in growth that the full model explains, which is the relevant quantity when asking how much of growth the regression actually predicts. The fact that both R^2 values are so low reflects the noisiness of the problem: Economic growth is extremely difficult to predict. The minimal incremental gain from including climate variables suggests that their predictive contribution is negligible. Of course, in-sample R^2 is not a valid measure of generalization performance; proper validation requires out-of-sample prediction.

Next, we consider naïve i.i.d. 10-fold cross-validation. This procedure randomly assigns (region-year) observations to folds but ignores spatiotemporal dependence, thereby allowing substantial data leakage. For example, if most US states’ growth in 2009 is in-sample while a few are held out, predictions for the held-out states still benefit from highly correlated in-sample observations from the same year. Despite this flaw, we report the results of i.i.d. 10-fold cross-validation in the Supplementary Materials (fig. S10). We find that models including climate variables perform no better than those that exclude them.

To address data leakage from temporal autocorrelation, McShane and Wyner (15) recommend “blocked” cross-validation using contiguous blocks of years. On our 38-year dataset, we form six blocks (four of 6 years and two of 7 years). Because year fixed effects cannot be applied to unseen years, we instead consider models without year effects or with random year effects that predict unseen years with zero mean. Although this departs from the original specification (Eq. 1), it enables a fair blocked cross-validation (see fig. S11). Again, we find no evidence that including climate variables improves predictive performance.

Last, to exploit the autoregressive prior on year effects in our Bayesian model (Eq. 3), we implement a rolling one-step-ahead prediction test. This is the prequential, or predictive-sequential, framework: At each point, a model is evaluated by its prediction of the next unseen outcomes using only the information available up to that point (17). Under the prequential view, the purpose of statistical inference is to make sequential probability forecasts for future observations, rather than to express information about parameters; hence, rolling one-step-ahead prediction is the natural way to assess whether the climate variables contribute stable information about future economic growth. At each step, the model is trained on years 1 through t and then used to predict year $t + 1$. Because $\alpha_{t+1}^{(\text{year})}$ is unknown at forecast time, we substitute the AR(1) forecast $\phi\alpha_t^{(\text{year})}$. We run this procedure for 20 years, from $t = 15$ (1995) through $t = 34$ (2014). We compare three models: (i) the overall mean $\bar{g}$ of the training data, (ii) the Bayesian model (Eq. 3) with climate variables η_{it} , and (iii) an otherwise identical Bayesian model without climate variables. Figure 8 reports the results. Both Bayesian models sometimes outperform the overall mean, but the gains are modest, inconsistent, and the median reduction in error is essentially zero. This reflects the substantial noise inherent in forecasting economic growth. Most crucially, the two Bayesian models perform nearly identically, providing no evidence that including climate variables improves predictive accuracy.

Note that the lag-growth Bayesian specification of the “Bayesian model” section yields the same out-of-sample picture: Including climate variables does not improve predictive accuracy under blocked cross-validation or the rolling one-step-ahead test, and its posterior predictive checks reproduce country- and region-level mean growth in the same way as the AR(1) year-effect model. The two Bayesian specifications thus disagree about the sign of the headline marginal effect but agree on the broader diagnostic picture. To summarize: Weak out-of-sample performance shows that the C&M regression is not useful as a forecasting model for region-year growth. This does not, by itself, prove that the estimated coefficients are noncausal; it is another piece of evidence, in combination with the previous analyses in this paper, that the original specification extracts much less stable independent climate-growth signal than its reported SEs imply.

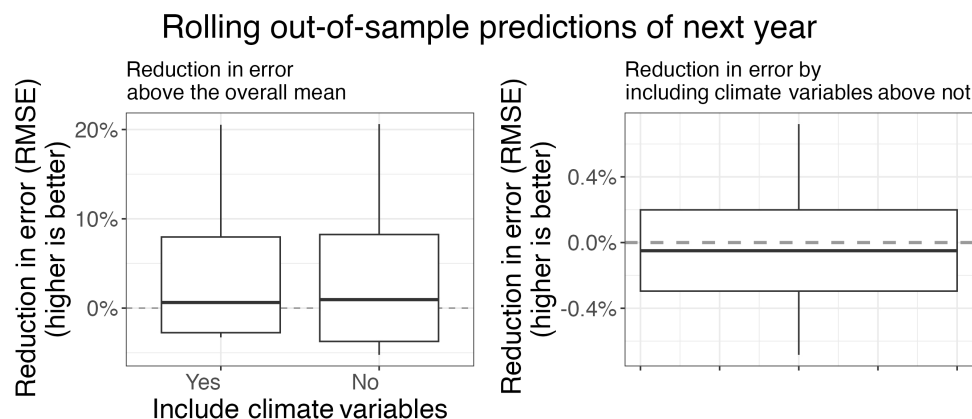


Fig. 8. Out-of-sample prediction comparison. For each of the 20 out-of-sample years $t + 1$, we compute the posterior mean RMSE for three models fit on years 1 through t : the overall mean $\bar{g}$ of the training data, the Bayesian model (3) with climate variables η_{it} , and an otherwise identical model without them. Left: reduction in RMSE relative to the overall mean for the Bayesian models with (“yes”) and without (“no”) climate variables. Right: reduction in RMSE of the model with climate variables relative to the model without them.

Supplementary Materials

This PDF file includes:

Sections S1 to S4

Figs. S1 to S11

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